

Examining correlation of sim data

Setup

```
library(here)
library(arrow)
library(Hmisc)
library(tidyverse)
library(data.table)
library(patchwork)
```

Load data

```
sim_data <- read_parquet(here("2-hosp", "sim-data", "h0_forward.parquet")) |>
  rename(
    id = pat_id,
    time = week)
real_data <- read_parquet(here("data", "hosp_long_weeks_two_lags.parquet")) |>
  filter(week > 0) |>
  rename(
    id = pat_id,
    time = week)
```

Carry Death Forward

```
df_state_5 <-
  real_data |>
  group_by(id) |>
  mutate(
    y = as.integer(y),
    yprev = as.integer(yprev),
    death_time = ifelse(any(y == 5), time[y == 5], NA_real_)
  ) |>
  filter(y == 5) |>
  complete(time = death_time:26, fill = list(y = 5L, yprev = 5L)) |>
  arrange(id, time) |>
  fill(everything())

df_state_1_to_4 <-
  real_data |>
  mutate(y = as.integer(y), yprev = as.integer(yprev)) |>
  filter(y != 5) |>
  arrange(id, time)

real_forward <-
  bind_rows(df_state_1_to_4, df_state_5) |>
  mutate(y = as.ordered(y), yprev = as.factor(yprev)) |>
  arrange(id, time)
```

Examine Correlation

Real data

```
corr_actual <-  
  real_forward |>  
  select(id, time, y) |>  
  arrange(time) |>  
  mutate(y = as.integer(y)) |>  
  pivot_wider(names_from = time, values_from = y) |>  
  ungroup() |>  
  select(-id) |>  
  cor(method="spearman", use="pairwise.complete.obs")  
  
p_corr_actual <- corr_actual |>  
  as.data.frame() |>  
  rownames_to_column() |>  
  pivot_longer(-rowname) |>  
  mutate(rowname = as.integer(rowname), name = as.integer(name)) |>  
  filter(rowname < name) |> #remove tiles below diagonal, because redundant  
  mutate(rowname = as.factor(rowname), name = as.factor(name)) |>  
  ggplot() +  
    aes(x=rowname, y=name, fill=value) +  
    geom_tile() +  
    geom_text(mapping=aes(x=rowname, y=name, label=round(value, 2)),  
              color="black",  
              size = 2.5) +  
    scale_fill_viridis_c(limits=c(-0.01,1), n.breaks = 10) +  
    expand_limits(value = c(0, 1)) +  
    # cowplot::theme_half_open(font_size=FONT_SIZE,  
    #                           font_family = "Source Sans Pro") +  
    theme(plot.title = element_text(hjust = 0.5),  
          panel.grid.major = element_blank(),  
          panel.grid.minor = element_blank()) +  
    theme(axis.line = element_blank(),  
          axis.ticks = element_blank()) +  
    theme(legend.key.height = unit(0.8, "in")) +  
    labs(x="Study week",  
         y="Study week",  
         fill="",  
         subtitle = "Actual Data")  
  
ggsave(p_corr_actual, filename = here("img", "fig-emp-cor-heat-hosp-weekly.svg"),  
        height = 10, width = 11)
```

Simulated Data based on model

```
corr_sim <-  
  sim_data |>  
  select(id, time, y) |>  
  arrange(time) |>  
  mutate(y = as.integer(y)) |>  
  pivot_wider(names_from = time, values_from = y) |>  
  ungroup() |>  
  select(-id) |>
```

```

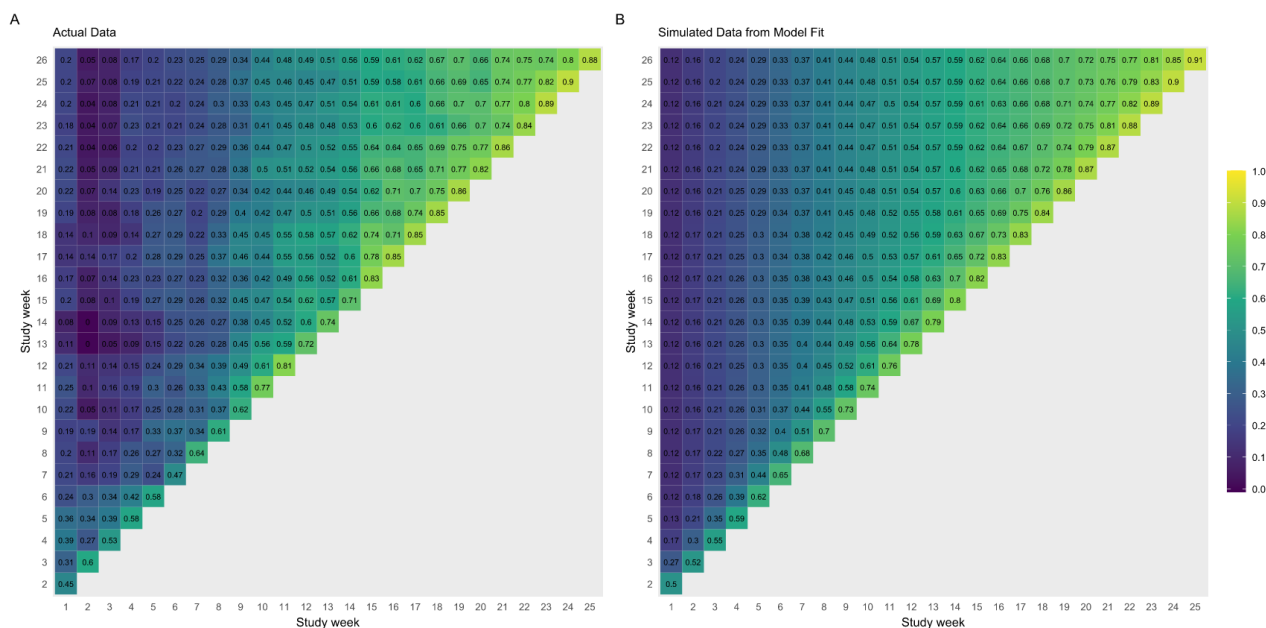
cor(method="spearman", use="pairwise.complete.obs")

p_corr_sim <- corr_sim |>
  as.data.frame() |>
  rownames_to_column() |>
  pivot_longer(-rowname) |>
  mutate(rowname = as.integer(rowname), name = as.integer(name)) |>
  filter(rowname < name) |> #remove tiles below diagonal, because redundant
  mutate(rowname = as.factor(rowname), name = as.factor(name)) |>
  ggplot() +
    aes(x=rowname, y=name, fill=value) +
    geom_tile() +
    geom_text(mapping=aes(x=rowname, y=name, label=round(value, 2)),
              color="black",
              size = 2.5) +
    scale_fill_viridis_c(limits=c(-0.01,1), n.breaks = 10) +
    expand_limits(value = c(0, 1)) +
    # cowplot::theme_half_open(font_size=FONT_SIZE,
    #                           font_family = "Source Sans Pro") +
    theme(plot.title = element_text(hjust = 0.5),
          panel.grid.major = element_blank(),
          panel.grid.minor = element_blank()) +
    theme(axis.line = element_blank(),
          axis.ticks = element_blank()) +
    theme(legend.key.height = unit(0.8, "in")) +
    labs(x="Study week",
         y="Study week",
         fill="",
         subtitle = "Simulated Data from Model Fit")

ggsave(p_corr_sim, filename = here("img", "fig-sim-cor-heat.svg"), height = 10, width =
11)

p_corr_actual + p_corr_sim + plot_layout(guides = "collect") +
  plot_annotation(tag_levels = 'A')

```



```
ggsave(filename = here("img", "fig-sim-cor-heat-comparison.svg"), height = 8, width = 16)
```

Absolute differences between correlation matrices

Means of absolute differences, stratified by the day involved.

```
ad <- abs(corr_actual - corr_sim)
m <- round(apply(ad, 1, mean), 2) |> data.table() |>
  rename(abs_diff = V1)

m |>
  rownames_to_column("rownames") |>
  pivot_longer(names_to = "model", values_to = "value", -rownames) |>
  pivot_wider(names_from = rownames, values_from = value) |>
  select(-model) |>
  knitr::kable()
```

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26
0.08	0.08	0.09	0.06	0.07	0.09	0.11	0.10	0.07	0.06	0.05	0.05	0.08	0.08	0.05	0.05	0.04	0.04	0.05	0.06	0.05	0.05	0.07	0.05	0.06	0.06	

```
m |> summarise_all(sum)
```

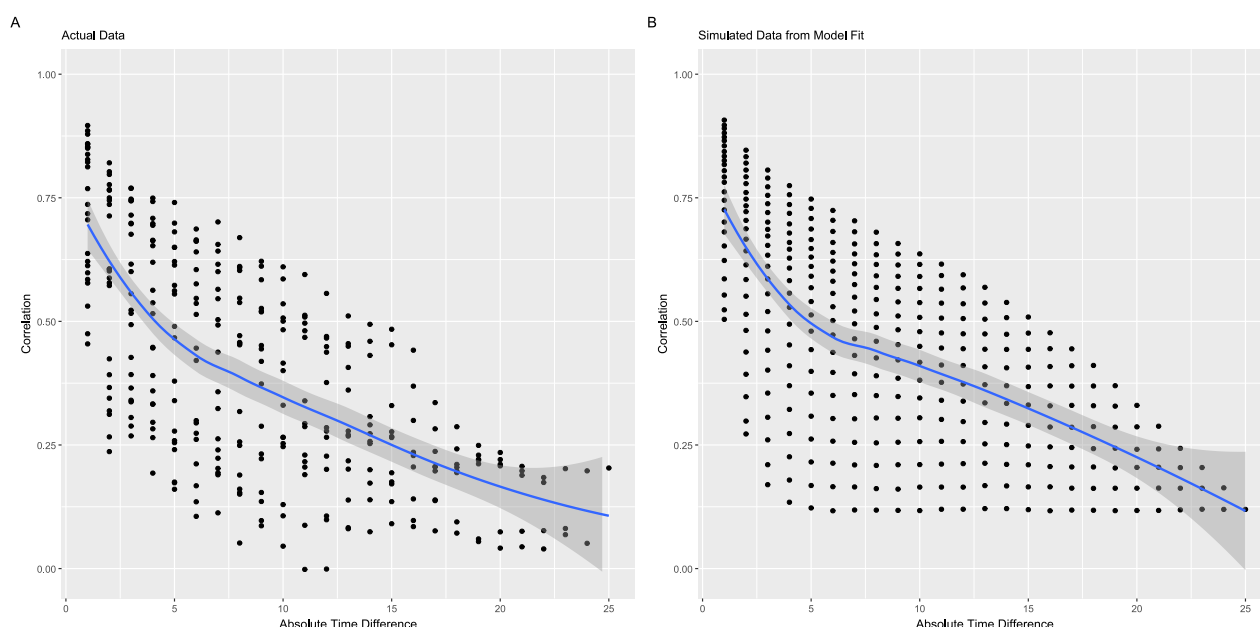
```
abs_diff
1      1.7
```

```
m |> summarise_all(mean)
```

```
abs_diff
1 0.06538462
```

Variograms

```
var1 <- Hmisc::plotCorrM(corr_actual)[[2]] +  
  scale_y_continuous(limits = c(-.01, 1)) +  
  labs(subtitle = "Actual Data")  
  
ggsave(filename = here("img", "fig-hosp-weekly-variogram-emp.svg"), height = 10, width  
= 11)  
  
var2 <- Hmisc::plotCorrM(corr_sim)[[2]] +  
  scale_y_continuous(limits = c(-.01, 1)) +  
  labs(subtitle = "Simulated Data from Model Fit")  
  
ggsave(filename = here("img", "fig-hosp-weekly-variogram-no-int.svg"), height = 10,  
width = 11)  
  
var1 + var2 + plot_annotation(tag_levels = 'A')
```



```
ggsave(filename = here("img", "fig-hosp-weekly-variogram-comparison.svg"), height = 8,  
width = 16)
```